



Capital University of Science and Technology

Department of Computer Science

CS3723 – Database Security

Semester Project

Semester: Spring 2026

Max Marks: 10

Instructor: Snober Naseer

Date Assigned: 6th June, 2026

Report Submission Deadline: 18th June 2026

Instructions:

Students may work individually or in instructor-approved groups. Every project must include database design, security implementation, attack simulation or risk assessment, testing, and evaluation. The selected project should demonstrate practical application of database security concepts such as access control, auditing, privacy preservation, monitoring, encryption, attack detection, or compliance

Learning Objectives:

By completing this project, you will:

- Develop a solid understanding of fundamental Concept of Database security.
- Explore practical applications of database security in various domains.
- Apply theoretical knowledge through hands-on development and analysis.

Optional Tools/Software:

- Microsoft Word (for documentation)
- PDF format (for final submission)

AI Project Topics (Choose One):

1. Database Intrusion Detection System (DIDS)
2. Database Activity Monitoring System (DAM)
3. Database Firewall for SQL Query Filtering
4. Privacy-Preserving Healthcare Database
5. Database Honeypot and Attack Analysis Platform
6. Insider Threat Detection Framework
7. Secure Banking Database with Fraud Analytics
8. Zero Trust Database Security Architecture

9. Blockchain-Based Audit Logging System
10. Differential Privacy Analytics Engine
11. Secure Multi-Tenant SaaS Database
12. Database Vulnerability Assessment Tool
13. Data Loss Prevention System for Databases
14. Secure Research Data Sharing Platform
15. Database Forensics Investigation System
16. Role-Based and Attribute-Based Access Control System
17. Cloud Database Security Management Platform
18. Database Ransomware Recovery Framework
19. Encrypted Database Search Engine
20. Enterprise Privilege Management System
21. Secure University Information System
22. Healthcare Compliance Monitoring Platform
23. Database Threat Intelligence Dashboard
24. Automated Database Risk Assessment System
25. Secure Electronic Voting Database
26. Database Security Operations Center (DB-SOC)
27. AI-Based SQL Attack Detection System
28. Data Anonymization and Re-identification Laboratory
29. Financial Transaction Monitoring Database
30. Enterprise Database Compliance Auditor

Submission Requirements

Each group must submit a complete project package consisting of a technical report, source code, database scripts, screenshots, testing evidence, and presentation slides. The technical report should include the title page, problem statement, objectives, literature review, system architecture, database design, security implementation, testing methodology, results, discussion, limitations, and conclusion.

Project Demonstration Requirements

Students must demonstrate the working system during the final evaluation. The demonstration should clearly show implemented security controls, attack scenarios, mitigation techniques, and system outputs. Students should be prepared to answer technical questions related to architecture, implementation, and security design decisions.

Academic Integrity Policy

Plagiarism, copied code, reused reports, or duplicated projects are strictly prohibited. Students must understand and explain every component of their implementation. Any project found to be copied may receive zero marks.

Project Report Structure

Title Page
Abstract
Introduction
Problem Statement
Objectives
System Design
Database Design
Security Controls Implemented
Screenshots

Testing and Evaluation
Results and Discussion
Conclusion
References

- Be ready for a **demo or viva** on your selected project.

Good Luck



